A FLAT FINITE MODULE OVER A HENSELIAN LOCAL BASE IS A FILTERED COLIMIT OF FREE MODULES WITH INJECTIVE REDUCTIONS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Projective neighbourhoods of a finite module flat at a point, in the henselian case. Same chapter flat.tex, label lemma-flat-finite-type-affine-neighbourhood-projective, invoked at proof line 5779. Gives a fundamental system of affine open neighbourhoods of the point on which the global sections of the module are free over the base, which produces the filtered system of free modules.
- Universal injectivity of the transition maps, and the Mittag-Leffler criterion for a colimit of universally injective maps. Same chapter flat.tex, label lemma-universally-injective-local (proof line 5791); Algebra, label algebra-lemma-colimit-universally-injective-ML (proof line 5794). The first shows that the structure maps of the system are universally injective over the base, and the second converts that into the Mittag-Leffler property of the colimit.

Source reference. Original lemma 0AS7: sources/stacks-chow/tags.txt, source lines 13570–13570. Navigation pointer for \ref{algebra-lemma-colimit-universally-injective-ML} used inside the counted span (line 5794); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MF: sources/stacks-flat/pinned-original-source.tex, source lines 5728–5728. Navigation pointer for \ref{lemma-flat-finite-type-affine-neighbourhood-projective} used inside the counted span (line 5779); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 05FQ: sources/stacks-flat/pinned-original-source.tex, source lines 1946–1946. Navigation pointer for \ref{lemma-universally-injective-local} used inside the counted span (line 5791); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $A \to B$ be a local ring map of local rings which is essentially of finite type. Let N be a finite B-module which is flat as an A-module. If A is henselian, then N is a filtered colimit*

$$N = \operatorname{colim}_i F_i$$

of free A-modules F_i such that all transition maps $u_i : F_i \to F_{i'}$ of the system induce injective maps $\overline{u}_i : F_i/\mathfrak{m}_A F_i \to F_{i'}/\mathfrak{m}_A F_{i'}$. Also, N is a Mittag-Leffler A-module.

Proof. We can find a morphism of finite type $X \to S = \operatorname{Spec}(A)$ and a point $x \in X$ lying over the closed point s of S and a finite type quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ such that $\mathcal{F}_x \cong N$ as an A-module. After shrinking X we may assume that each point of $\operatorname{Ass}_{X_s}(\mathcal{F}_s)$ specializes to x. By Lemma 05MF we see that there exists a fundamental system of affine open neighbourhoods $U_i \subset X$ of x such that $\Gamma(U_i, \mathcal{F})$ is a free A-module F_i. Note that if $U_{i'} \subset U_i$, then

$$F_i/\mathfrak{m}_A F_i = \Gamma(U_{i,s}, \mathcal{F}_s) \longrightarrow \Gamma(U_{i',s}, \mathcal{F}_s) = F_{i'}/\mathfrak{m}_A F_{i'}$$

is injective because a section of the kernel would be supported at a closed subset of X_s not meeting x which is a contradiction to our choice of X above. Since the maps $F_i \to F_{i'}$ are A-universally injective (Lemma 05FQ) it follows that N is Mittag-Leffler by Algebra, Lemma 0AS7. $\square$